Data Sources

Data availability and provenance statement for "Dynamic Joint Calibration of Combined VaR and ES Forecasts: Evidence from US Industries and the Yen"

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This document describes the data used in the paper: where each data set comes from, when it was downloaded, which part of it is used, how it is processed and whether it is included in the replication package. Its structure follows the template README of the Social Science Data Editors (Vilhuber et al., 2022). The package is available at https://github.com/sheyung21-crypto/var-es-dynamic-joint-calibration.

1. Statement about Rights

The author obtained all data from public websites and had legitimate access to them for this research. Only one data set, the yen exchange rate from FRED, may be redistributed: FRED marks it "Public Domain: Citation Requested", and the Federal Reserve Board permits copying of its information and asks for attribution (https://www.federalreserve.gov/disclaimer.htm). Permission to redistribute the Kenneth R. French data and the Yahoo Finance data could not be confirmed, so these files are not included in the package. Their download addresses, file sizes and SHA-256 hashes are recorded instead, so that a user who obtains the same files can verify them.

2. Summary of Availability

All data come from public providers, but only part of them can be included in the package. Table 1 summarizes the sources.

Table 1. Availability of the data

Data	Provider	Used in	Included in the package
Daily returns of 30 US industry portfolios	Kenneth R. French Data Library	Study 1 (main sample)	No
Daily closing prices of six equity indices	Yahoo Finance	Study 1 (secondary sample)	No
Daily yen–US dollar exchange rate (DEXJPUS), 1974–2009	FRED, Federal Reserve Bank of St. Louis	Study 2	Yes
Five daily series used to set the parameters of the method	FRED	Development only	No

Note: "Development only" means that the data were used to choose the settings of the method before the evaluation. They are not needed to reproduce the reported results.

Without the Study 1 files, a user can still rerun Study 2, check the archived results of Study 1 and rebuild all tables and figures of the paper.

3. Details on Each Data Source

3.1 US industry portfolios (Study 1)

The main sample consists of the daily value-weighted returns of the 30 industry portfolios in the Kenneth R. French Data Library, Tuck School of Business at Dartmouth (French, 2026). The portfolios are constructed from CRSP data; the author did not use the CRSP database itself. The file `30_Industry_Portfolios_daily_CSV.zip` was downloaded on 23 October 2026 from https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/ftp/30_Industry_Portfolios_daily_CSV.zip. Its header identifies the 202608 CRSP database. The construction of the portfolios is described at https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/Data_Library/det_30_ind_port.html.

Table 2. US industry portfolios: data used

Item	Description
Part of the file	Average Value Weighted Returns -- Daily
Industries	30: Food, Beer, Smoke, Games, Books, Hshld, Clths, Hlth, Chems, Txtls, Cnstr, Steel, FabPr, ElcEq, Autos, Carry, Mines, Coal, Oil, Util, Telcm, Servs, BusEq, Paper, Trans, Whlsl, Rtail, Meals, Fin, Other
Sample period	1 July 1963 – 31 August 2026; 15,897 daily returns per industry
Evaluation period	3 July 1975 – 31 August 2026; 12,897 trading days, the same for all industries
Units	Percentage simple returns r_t
Transformation	Log returns $y_t = 100 \log(1 + r_t/100)$
Missing values	The codes −99.99 and −999 are removed; none occurs in the sample period

Source: Kenneth R. French Data Library; `data/sample_inventory.csv`.

Other parts of the file, such as the equal-weighted returns and the number of firms, are not used. The returns are not winsorized or adjusted in any other way.

3.2 International equity indices (Study 1)

The secondary sample consists of six equity indices: the Taiwan Weighted Index, the KOSPI, the FTSE 100, the DAX, the S&P/ASX 200 and the S&P/TSX Composite (Yahoo Finance, 2026). Daily data were downloaded from the Yahoo Finance chart service in JSON format on 23 October 2026. The analysis uses the closing price; the adjusted closing price, the opening, high and low prices and the trading volume are not used. Table 3 lists the sample of each index.

Table 3. International equity indices: samples and data cleaning

Symbol	Return sample	Removed closes	Returns	Evaluation period	Evaluation days
^TWII	3 Jul 1997 – 22 Oct 2026	142	7,181	14 Sep 2009 – 22 Oct 2026	4,181
^KS11	12 Dec 1996 – 22 Oct 2026	150	7,348	24 Feb 2009 – 22 Oct 2026	4,348
^FTSE	4 Jan 1984 – 23 Oct 2026	183	10,814	16 Nov 1995 – 23 Oct 2026	7,814
^GDAXI	4 Jan 1988 – 22 Oct 2026	169	9,815	21 Dec 1999 – 22 Oct 2026	6,815
^AXJO	24 Nov 1992 – 23 Oct 2026	101	8,575	30 Sep 2004 – 23 Oct 2026	5,575
^GSPTSE	3 Jul 1979 – 23 Oct 2026	250	11,880	25 Jun 1991 – 23 Oct 2026	8,880

Note: The symbols are listed in the same order as the index names in the text. Removed closes are missing or nonpositive closing prices. One duplicate date in ^AXJO is resolved by keeping the last record. Together, the six evaluation periods cover 9,119 dates from 25 June 1991 to 23 October 2026; on each date, results are averaged over the indices with data. Source: Yahoo Finance; `data/sample_inventory.csv`.

Dates are taken from the time stamps in the files and adjusted by the time-zone offset that the files record. Returns are computed as $y_t = 100 \log(P_t/P_{t-1})$ between consecutive available closing prices. Missing trading days are not filled.

According to Yahoo, downloading historical data as a file requires a paid subscription, and its terms restrict automated collection (https://help.yahoo.com/kb/sln2311.html; https://legal.yahoo.com/us/en/yahoo/terms/otos/index.html). A new download may also differ from the files used in the paper. The exact requests are recorded in `data/study1/expected_fetch_manifest.json`.

3.3 Yen exchange rate (Study 2)

Study 2 uses the daily yen–US dollar exchange rate, series DEXJPUS (Board of Governors of the Federal Reserve System, 2026a). The data come from the H.10 release of the Federal Reserve Board and were downloaded from FRED, Federal Reserve Bank of St. Louis, on 24 October 2026. The rate is quoted in yen per US dollar and is not seasonally adjusted, so a negative return means that the yen appreciates. The file is included in the package as `data/study2/DEXJPUS_1974_2009.csv`.

The requested period, 2 January 1974 to 31 December 2009, contains 9,392 dates. After 348 dates without a value are removed, 9,044 rates and 9,043 returns remain. Table 4 shows how the returns are used.

Table 4. Yen exchange rate: use of the returns

Returns	Dates	Use in the study
1–1,000	3 January 1974 – 29 December 1977	Estimation of the base models
1,001–2,000	from 30 December 1977	Forecasts build the score history needed for MCS selection
2,001–3,000	from 29 December 1981	Calibration starts; warm-up period
3,001–9,043	24 December 1985 – 31 December 2009	Evaluation period: 6,043 trading days

Note: Returns are $y_t = 100 \log(P_t/P_{t-1})$ between consecutive available rates. Source: FRED; `data/sample_inventory.csv`.

3.4 Development data (not used for the reported results)

Before the evaluation, the method was developed on other data. The final development rounds used the five daily FRED series in Table 5. They were used only to choose the settings of the method; in particular, the bootstrap block lengths were computed from score differences on the S&P 500, the Dow Jones Industrial Average and the NASDAQ Composite. The paper (Section 4.1) also notes that international equity indices from earlier work were inspected during development; no fixed setting was identified as coming from them. None of these data is needed to reproduce the reported results. The equity series are not included because their providers require permission for reproduction, and the exchange-rate series are not needed.

Table 5. Development data

FRED series	Description	Original provider	Daily data used
SP500	S&P 500	S&P Dow Jones Indices LLC	24 Oct 2016 – 21 Oct 2026
DJIA	Dow Jones Industrial Average	S&P Dow Jones Indices LLC	24 Oct 2016 – 21 Oct 2026
NASDAQCOM	NASDAQ Composite	Nasdaq, Inc.	4 Jan 2016 – 21 Oct 2026
DEXUSEU	US dollars per euro	Federal Reserve Board (H.10)	4 Jan 2010 – 16 Oct 2026
DEXJPUS	Yen per US dollar	Federal Reserve Board (H.10)	4 Jan 2010 – 16 Oct 2026

Source: FRED, downloaded on 22 October 2026; `provenance/Development_Provenance.md`.

4. Dataset List

Table 6 lists the input files and the records that describe them. Files that are not included must be placed in the folder structure given in `data/ACCESS.md`.

Table 6. Data files

File	Source	Included	Content
`data/study2/DEXJPUS_1974_2009.csv`	FRED	Yes	Yen–US dollar rate, 1974–2009
`french/30_Industry_Portfolios_daily_CSV.zip`	French Data Library	No	30 industry portfolios, daily
`yahoo/yahoo_TWII.json` and five similar files	Yahoo Finance	No	Daily data of the six indices
`data/study1/expected_fetch_manifest.json`	Author	Yes	Addresses, sizes, download times and hashes of the seven Study 1 files
`data/study2/fetch_manifest.json`	Author	Yes	Download record of the yen file
`data/sample_inventory.csv`	Author	Yes	Sample dates and counts of all 37 series
`data/ACCESS.md`	Author	Yes	How to obtain and check the Study 1 files

5. Hashes of the Input Files

The SHA-256 hash identifies the content of a file exactly. A file obtained later must have the hash in Table 7 to reproduce the paper exactly. The program `code/data/manage_inputs.py` checks the hashes before any estimation.

Table 7. Sizes and SHA-256 hashes of the eight input files

File	Bytes	SHA-256
`30_Industry_Portfolios_daily_CSV.zip`	2,597,362	`04a7a74fcd7c12e90461718e5ba8490e99d18c3e9323182b6c7a7b320e6cd34a`
`yahoo_TWII.json`	689,915	`0ca89944534f38a8b298c0899a73df02fb37953895797a6c7ae80f1ab8b93329`
`yahoo_KS11.json`	771,428	`03d456451d3fa3b068044c8d29b9a1046db7c2f5152f8a8d2accfd57b5167b0f`
`yahoo_FTSE.json`	1,010,771	`a8e62f7fb39e9c68fbe5c9e96f638c8aa8461fdf90ebff436dadb183a6ae0edf`
`yahoo_GDAXI.json`	975,505	`bb64d171189e255c12748f55c34f1894b70b5e0426327bb29b44b166b5d7ce60`
`yahoo_AXJO.json`	769,445	`20921f149b40432bcb20baa4fd401bd339a6a375565a27b56664c2ed1c6b7e10`
`yahoo_GSPTSE.json`	1,117,430	`704065bd537bedd65a0a4fe60bce6bf81b3923d6bf86e30b70f7dcfbae52b519`
`DEXJPUS_1974_2009.csv`	166,423	`90fa5f9af7dd2ba1754958b11114e41f48946efa440f8afc9f62b50a09a479b3`

6. Remarks

1. All series are the versions offered by the providers in October 2026. Past values may have been revised, so the data are not necessarily what was known at each past date.
2. The variables analyzed in the paper, such as the VaR and ES forecasts, the violation indicators, the FZ0 scores and the test statistics, are computed by the code. They are not separate data sources.

7. Data Citations

Board of Governors of the Federal Reserve System (US) (2026a). Japanese Yen to U.S. Dollar Spot Exchange Rate [DEXJPUS]. Retrieved from FRED, Federal Reserve Bank of St. Louis (https://fred.stlouisfed.org/series/DEXJPUS, accessed 24 October 2026).

Board of Governors of the Federal Reserve System (US) (2026b). U.S. Dollars to Euro Spot Exchange Rate [DEXUSEU]. Retrieved from FRED, Federal Reserve Bank of St. Louis (https://fred.stlouisfed.org/series/DEXUSEU, accessed 22 October 2026).

French, K. R. (2026). 30 Industry Portfolios, daily. Kenneth R. French Data Library, Tuck School of Business at Dartmouth (https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html, accessed 23 October 2026).

Nasdaq, Inc. (2026). NASDAQ Composite [NASDAQCOM]. Retrieved from FRED, Federal Reserve Bank of St. Louis (https://fred.stlouisfed.org/series/NASDAQCOM, accessed 22 October 2026).

S&P Dow Jones Indices LLC (2026a). Dow Jones Industrial Average [DJIA]. Retrieved from FRED, Federal Reserve Bank of St. Louis (https://fred.stlouisfed.org/series/DJIA, accessed 22 October 2026).

S&P Dow Jones Indices LLC (2026b). S&P 500 [SP500]. Retrieved from FRED, Federal Reserve Bank of St. Louis (https://fred.stlouisfed.org/series/SP500, accessed 22 October 2026).

Vilhuber, L., Connolly, M., Koren, M., Llull, J., and Morrow, P. (2022). A template README for social science replication packages, version 1.1. Social Science Data Editors (https://doi.org/10.5281/zenodo.7293838).

Yahoo Finance (2026). Daily data for ^TWII, ^KS11, ^FTSE, ^GDAXI, ^AXJO and ^GSPTSE (https://finance.yahoo.com, accessed 23 October 2026).